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WM-Concord: Do World Model Diagnostics Agree?

A Cross Diagnostic Validity Study of World Model Diagnostics

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Abstract

Sequence models are often said to learn a “world model”: an internal picture of the state of the domain they were trained on. The literature tests this claim with four families of diagnostics: next token validity, state equivalence metrics based on the Myhill Nerode theorem, inductive bias probes, and representation probing. These diagnostics have never been run on the same models, and on Othello they already reach opposite conclusions. WM-Concord runs all four diagnostics on one controlled zoo of small sequence models trained on three domains with a known ground truth automaton (navigation, Othello, and a program interpreter). We ask whether the diagnostics rank models the same way, why they disagree when they do, and which one best predicts failure on related but shifted tasks.

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1. Introduction

Whether a sequence model trained only to predict the next token builds a coherent internal model of its domain is one of the central open questions in current machine learning. Several research groups have built diagnostics to answer it, but each group uses its own instrument, on its own models, in its own domain. The same Othello models have been reported both as having a world model [1, 2] and as lacking a coherent one [3, 4]. This project does not propose a new diagnostic. It measures whether the existing ones agree, and which one can be trusted.

Contributions.

- The first head to head comparison of four world model diagnostics on the same models and the same ground truth.
- A test of which diagnostic predicts failure on related but shifted tasks, judged against a fragility suite that is frozen before any diagnostic is run.
- An open source benchmark (WM-Concord) with one common interface, so a future diagnostic can be compared against these four on identical ground truth.

[To be written: one paragraph summarising what we found, written last]

2. Problem Statement

The four diagnostic families studied in this project are:

D1, next token validity. The fraction of the model’s proposed next tokens that are legal in the true state.

D2, state equivalence. Compression error (two histories that reach the same state are treated differently) and distinction error (two different states are treated the same), following [3].

D3, inductive bias probes. How the model adapts to small synthetic tasks generated from the true world model [4].

D4, representation probing. Whether the true state can be read from the model’s internal activations by a probe, checked against a control task and by causal intervention [1].

We test four hypotheses:

H1, concordance. Do the four diagnostics rank the same models in the same order?

H2, disagreement. When they disagree, is the disagreement systematic (tied to domain, architecture, scale, or training data) or just noise?

H3, predictive validity. Which diagnostic best predicts failure on a frozen suite of related but shifted tasks?

H4, generalization. Do the answers hold on a held out domain and a held out architecture?

Challenges. The diagnostics measure different things (behaviour, adaptation, and internal activations), so their scores are not on a common scale. The state equivalence metrics cannot be computed exactly in large domains and must be estimated by sampling. Probes can report structure the model does not actually use. The design handles each of these in Section 5.

3. Related Work

3.1. Representation probing

Li et al. [1] trained a GPT model on Othello move sequences only. The model never sees a board or the rules, yet it proposes legal moves almost every time, and probes trained on its internal activations recover the board state. They then edited the decoded board inside the model and showed that its next move predictions changed to match, which is evidence that the representation is used and not only present. Yuan and Søgaard [2] extended this to seven language models fine tuned on Othello and found similar board representations across all of them, which they read as support for a real world model. Both works use probing as the only instrument. WM-Concord uses probing as one of four diagnostics, adds a control task and a causal check, and compares its verdict with the other three on the same models.

[To be written: Nanda et al. 2023 and Toshniwal et al. 2022, one paragraph each with how WM-Concord differs]

3.2. State equivalence metrics

Vafa et al. [3] define the domain as a deterministic finite automaton and derive two metrics from the Myhill Nerode theorem. Compression error measures whether the model gives the same continuations to two histories that end in the same state; distinction error measures whether it can tell apart histories that end in different states. On taxi routes in Manhattan, Othello, and logic puzzles, models that looked good on next token accuracy failed these metrics, and a map reconstructed from the navigation model contained streets that do not exist. Models trained on random walks built more coherent world models than models trained on shortest paths. That work compares its metrics only with next token accuracy. WM-Concord uses them as diagnostic D2, adopts their three training distributions, and compares them with all three other diagnostics and with an external fragility criterion.

3.3. Inductive bias probes

Vafa et al. [4] ask a different question: does a model’s inductive bias point at the true world model? They adapt a pretrained model to small synthetic datasets generated from a postulated world model and test whether it extrapolates the way that world model would. A model trained on orbital trajectories predicted them well, but symbolic regression on the adapted model recovered force laws that make no physical sense. This is our diagnostic D3. WM-Concord applies it to the same zoo as the other diagnostics, so its verdict can be compared model by model.

3.4. Probing confounds

[To be written: Hewitt and Liang 2019 (control tasks and selectivity) and Belinkov 2022 (probing survey), one paragraph each, explaining why raw probe accuracy can overstate what a model computes and how WM-Concord guards against it]

3.5. Measurement validity in ML evaluation

[To be written: Jacobs and Wallach 2021 and Raji et al. 2021, one paragraph each, framing WM-Concord as a construct and predictive validity study of existing instruments]

3.6. How WM-Concord differs

These papers reach opposite conclusions about the same domain, but they use different instruments on different models, so their disagreement cannot be interpreted. WM-Concord runs all four instruments on one shared zoo of models with a known ground truth, which turns the

disagreement into something that can be measured (H1, H2) and judged against an external criterion (H3).

4. Data and Domains

All domains are defined by a deterministic finite automaton (DFA), so the true state is known at every step. Every domain emits the same triple: token sequence, true state, and the set of legal next tokens.

Domain	State	Training distributions
Navigation	current intersection on a street graph	shortest paths, noisy shortest paths, random walks
Othello	board position	synthetic, championship, mixed
Interpreter (new)	registers and stack of a small machine	random, short structured, long structured

Table 1: The three domains. Othello and interpreter distributions are tentative until their generators exist.

4.1. Navigation

[To be written: street graph source, DFA definition, the three samplers, state space size]

4.2. Othello

[To be written: game corpus (othello_world synthetic and championship games), board state, legal moves, DFA size]

4.3. Interpreter

[To be written: instruction set of the small stack or register machine, what counts as state, how DFA size is tuned]

4.4. Reproduction artifacts

[To be written: released checkpoints used to check our implementations: Vafa et al. 2024 navigation models (three splits) and the Othello GPT synthetic model]

5. Methodology

5.1. Model zoo

The zoo varies four axes on purpose, so that models differ in quality and the diagnostics have something to disagree about: 3 domains $\times$ 3 architectures (transformer, LSTM, state space model) $\times$ 3 scales $\times$ 3 training distributions $\times$ 3 seeds, for 243 models and 972 model and diagnostic evaluations.

[To be written: scale definitions (layers, width), training budget, validation threshold for admitting a model]

5.2. The four diagnostics

[To be written: one subsection each for D1 to D4: what it computes, the exact score written to the master table, and the sampling protocol. All four implement one common interface: `score(model, domain)` returns a dictionary]

5.3. Fragility suite

[To be written: per domain task variants (detour routing, altered Othello rule, changed instruction semantics), written as explicit DFA changes and frozen with a git tag before any diagnostic is run]

5.4. Analysis plan

[To be written: H1: Kendall tau and Spearman per pair and Kendall W overall, with bootstrap intervals. H2: regression of disagreement on domain, architecture, scale, and distribution, against the seed noise floor. H3: regression of fragility on each diagnostic score. H4: repeat on the held out domain and architecture]

6. Experimental Setup

All experiments run on one AWS g5.2xlarge instance (one NVIDIA A10G with 24 GB, 8 vC-PUs, 32 GB RAM). Library versions are pinned in `requirements.txt`, every run records its configuration and seed, and every result can be traced to the commit that produced it. Code, configurations, and the run matrix are in the project repository.

[To be written: measured training throughput per model and total compute for the zoo]

7. Results

7.1. Reproduction of published results

[To be written: our numbers next to the published ones for each reproduced result, with the tolerance chosen before comparing]

7.2. Concordance (H1)

[To be written: pairwise rank correlation heatmap and overall Kendall W]

7.3. Disagreement (H2)

[To be written: where and why the diagnostics disagree, compared with the seed noise floor]

7.4. Predictive validity (H3)

[To be written: which diagnostic best predicts fragility, and whether next token validity predicts it at all]

7.5. Held out domain and architecture (H4)

[To be written: whether the findings hold on data kept out of development]

8. Discussion

[To be written: what each diagnostic licenses a researcher to claim, and which to use when]

8.1. Limitations

[To be written: estimator variance of the state equivalence metrics, choice of state abstraction for the interpreter, architectures that could not be patched, single GPU scale]

9. Conclusion and Future Work

[To be written: short answers to H1 to H4, and extensions to continuous state and probabilistic domains]

References

- [1] K. Li, A. K. Hopkins, D. Bau, F. Viégas, H. Pfister, and M. Wattenberg, “Emergent world representations: Exploring a sequence model trained on a synthetic task,” in *International Conference on Learning Representations (ICLR)*, 2023.
- [2] Y. Yuan and A. Søgaard, “Revisiting the Othello world model hypothesis,” in *ICLR 2025 Workshop on World Models*, 2025.
- [3] K. Vafa, J. Y. Chen, A. Rambachan, J. Kleinberg, and S. Mullainathan, “Evaluating the world model implicit in a generative model,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [4] K. Vafa, P. G. Chang, A. Rambachan, and S. Mullainathan, “What has a foundation model found? using inductive bias to probe for world models,” in *International Conference on Machine Learning (ICML)*, 2025.

A. Team Roles

Member	Responsibility
Venkatesh Nagarjuna	D2 state equivalence metrics and automata
Nikhil Obuleni	D3 inductive bias probes
Siddardha Reddy	D4 representation probing and causal checks
Viharika Vemparala	D1 next token validity

Table 2: Diagnostic ownership. Domains, the model zoo, the analysis, and the paper are shared.

B. Reproducibility

[To be written: repository link, the one command that runs everything, and where each figure is generated]